Assembly of the Drell–Yan contributions up to N^{3}LO

UV renormalisation per coupling piece

DY_P2B — notes_Larin/

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1 Notation

V-vertex labels (the coupling is stripped off and written in the "coupling" column):

label	V attached to	coupling
v, a	open quark line 1 (flavour q)	v_q, a_q
v_2, a_2	open quark line 2 (flavour Q, four-quark channels)	v_Q, a_Q
$v_{\rm PS}$, $a_{\rm PS}$	closed quark loop = **pure singlet (PS)**, summed over n_f flavours	$\sum_f v_f$, $\sum_f a_f$

The couplings are real numbers (tree-level electroweak couplings; the Z propagator, complex in the Breit–Wigner form, multiplies the whole hadronic tensor and is not part of them), so they factor out of every 2Re below. Lines are labelled *by line*, not by flavour (identical quarks: see `Larin.pdf`). PS exists only in loop amplitudes: RRR, RR, R, LO have none.

Pieces. $Y^{(j)}X^{(i)*}$ = amplitude with label Y at j loops times the complex conjugate of the amplitude with label X at i loops, summed over spins, colours, polarisations, couplings stripped off, contracted with the symmetric leptonic projector L_1 (parity-even pieces, table 1) or with the antisymmetric projector L_6 (parity-odd pieces, one vector and one axial label, table 2; leptonic coupling $2v_\ell a_\ell$ instead of $v_\ell^2 + a_\ell^2$).

γ_5. Same-line parity-even products: anticommuting γ_5, $aa = vv$: these rows are $v\,v^*$ times $v_q^2 + a_q^2$. Rows marked **(L)**, and all parity-odd rows: Larin for *every* axial vertex of the piece and of its counterterm pieces.

Conventions. $a = \alpha_s(\mu^2)/(2\pi)$, $d = 4 - 2\epsilon$, $\mu = \mu_0$; the bare L-loop amplitude (in terms of $\bar\alpha_s$) is multiplied by S_ϵ^{-L}, $S_\epsilon = (4\pi)^\epsilon e^{-\epsilon\gamma_E}$ (arXiv:2306.10170, eq. (5.1)).

$$\beta_0 = \tfrac{11}{6}C_A - \tfrac{2}{3}T_R n_f, \qquad \beta_1 = \tfrac{17}{6}C_A^2 - \tfrac{5}{3}C_A T_R n_f - C_F T_R n_f.$$

2 Parity-even pieces (L_1)

m = number of emitted partons. Rows of one contribution are separate coupling pieces, summed after multiplying by their couplings. "= 0": see section 4.

Table 1: Parity-even pieces, contracted with L_1.

channels contribution		coupling	renormalised
LO (a^0)			
$q\bar{q} \to V$	B	$v_q^2 + a_q^2$	$v^{(0)}\, v^{(0)*}$
NLO (a^1)			
$q\bar{q} \to Vg$, $qg \to Vq$	R	$v_q^2 + a_q^2$	$v^{(0)}\, v^{(0)*}$
$q\bar{q} \to V$	V	$v_q^2 + a_q^2$	$2\mathrm{Re}\left[v^{(1)}\, v^{(0)*}\right]$
NNLO (a^2)			
$q\bar{q} \to Vgg$, $qg \to Vqg$, $gg \to Vq\bar{q}$	RR	$v_q^2 + a_q^2$	$v^{(0)}\, v^{(0)*}$
$q\bar{q} \to VQ\bar{Q}$, $qQ \to VqQ$, $q\bar{Q} \to Vq\bar{Q}$, $q\bar{q} \to Vq\bar{q}$, $qq \to Vqq$	RR	$v_q^2 + a_q^2$	$v^{(0)}\, v^{(0)*}$
		$v_Q^2 + a_Q^2$	$v_2^{(0)}\, v_2^{(0)*}$
		$v_q v_Q$	$2\mathrm{Re}\left[v^{(0)}\, v_2^{(0)*}\right]$
		$a_q a_Q$ (L)	$2\mathrm{Re}\left[a^{(0)}\, a_2^{(0)*}\right]$
$q\bar{q} \to Vg$, $qg \to Vq$	RV	$v_q^2 + a_q^2$	$2\mathrm{Re}\left[v^{(1)}\, v^{(0)*} - \frac{\beta_0}{2\epsilon}\, v^{(0)}\, v^{(0)*}\right]$
		$v_q \sum v_f$	$= 0$
		$a_q \sum a_f$ (L)	$2\mathrm{Re}\left[a_{\mathrm{PS}}^{(1)}\, a^{(0)*}\right]$

Table 1: Parity-even pieces, contracted with L_1.

channels	contribution	coupling	renormalised
$q\bar{q} \to V$	VV	$v_q^2 + a_q^2$	$2\mathrm{Re}\left[v^{(2)}\, v^{(0)*} - \frac{\beta_0}{\epsilon}\, v^{(1)}\, v^{(0)*}\right] + v^{(1)}\, v^{(1)*}$
		$v_q \sum v_f$	$= 0$
		$a_q \sum a_f$ (L)	$2\mathrm{Re}\left[a_{\mathrm{PS}}^{(2)}\, a^{(0)*} + C_F T_R\left(\frac{3}{2\epsilon} + \frac{3}{4}\right) a^{(0)}\, a^{(0)*}\right]$
N^{3}LO (a^3)			
$q\bar{q} \to Vggg$ $qg \to Vqgg$ $gg \to Vq\bar{q}g$	RRR	$v_q^2 + a_q^2$	$v^{(0)}\, v^{(0)*}$
$q\bar{q} \to VQ\bar{Q}g$ $qg \to VqQ\bar{Q}$ $qQ \to VqQg$ $q\bar{Q} \to Vq\bar{Q}g$ $q\bar{q} \to Vq\bar{q}g$ $qq \to Vqqg$ $qg \to Vqq\bar{q}$	RRR	$v_q^2 + a_q^2$	$v^{(0)}\, v^{(0)*}$
		$v_Q^2 + a_Q^2$	$v_2^{(0)}\, v_2^{(0)*}$
		$v_q v_Q$	$2\mathrm{Re}\left[v^{(0)}\, v_2^{(0)*}\right]$
		$a_q a_Q$ (L)	$2\mathrm{Re}\left[a^{(0)}\, a_2^{(0)*}\right]$
$q\bar{q} \to Vgg$ $qg \to Vqg$ $gg \to Vq\bar{q}$	RRV	$v_q^2 + a_q^2$	$2\mathrm{Re}\left[v^{(1)}\, v^{(0)*} - \frac{\beta_0}{\epsilon}\, v^{(0)}\, v^{(0)*}\right]$
		$v_q \sum v_f$	$2\mathrm{Re}\left[v_{\mathrm{PS}}^{(1)}\, v^{(0)*}\right]$
		$a_q \sum a_f$ (L)	$2\mathrm{Re}\left[a_{\mathrm{PS}}^{(1)}\, a^{(0)*}\right]$

Table 1: Parity-even pieces, contracted with L_1.

channels	contribution	coupling	renormalised
$q\bar{q} \to VQ\bar{Q}$, $qQ \to VqQ$, $q\bar{Q} \to Vq\bar{Q}$, $q\bar{q} \to Vq\bar{q}$, $qq \to Vqq$	RRV	$v_q^2 + a_q^2$	$2\mathrm{Re}\left[v^{(1)}\, v^{(0)*} - \frac{\beta_0}{\epsilon}\, v^{(0)}\, v^{(0)*}\right]$
		$v_Q^2 + a_Q^2$	$2\mathrm{Re}\left[v_2^{(1)}\, v_2^{(0)*} - \frac{\beta_0}{\epsilon}\, v_2^{(0)}\, v_2^{(0)*}\right]$
		$v_q v_Q$	$2\mathrm{Re}\left[v^{(1)}\, v_2^{(0)*} - \frac{\beta_0}{\epsilon}\, v^{(0)}\, v_2^{(0)*}\right] + 2\mathrm{Re}\left[v_2^{(1)}\, v^{(0)*} - \frac{\beta_0}{\epsilon}\, v_2^{(0)}\, v^{(0)*}\right]$
		$a_q a_Q$ (L)	$2\mathrm{Re}\left[a^{(1)}\, a_2^{(0)*} - \left(\frac{\beta_0}{\epsilon} + 2C_F\right) a^{(0)}\, a_2^{(0)*}\right] + 2\mathrm{Re}\left[a_2^{(1)}\, a^{(0)*} - \left(\frac{\beta_0}{\epsilon} + 2C_F\right) a_2^{(0)}\, a^{(0)*}\right]$
		$v_q \sum v_f,\ v_Q \sum v_f$	$= 0$
		$a_q \sum a_f$ (L)	$2\mathrm{Re}\left[a_{\mathrm{PS}}^{(1)}\, a^{(0)*}\right]$
		$a_Q \sum a_f$ (L)	$2\mathrm{Re}\left[a_{\mathrm{PS}}^{(1)}\, a_2^{(0)*}\right]$
$q\bar{q} \to Vg$, $qg \to Vq$	RVV 2L×0L	$v_q^2 + a_q^2$	$2\mathrm{Re}\left[v^{(2)}\, v^{(0)*} - \frac{3\beta_0}{2\epsilon}\, v^{(1)}\, v^{(0)*} + \left(\frac{3\beta_0^2}{8\epsilon^2} - \frac{\beta_1}{4\epsilon}\right) v^{(0)}\, v^{(0)*}\right]$
		$v_q \sum v_f$	$2\mathrm{Re}\left[v_{\mathrm{PS}}^{(2)}\, v^{(0)*}\right]$
		$a_q \sum a_f$ (L)	$2\mathrm{Re}\left[a_{\mathrm{PS}}^{(2)}\, a^{(0)*} - \left(\frac{3\beta_0}{2\epsilon} + 2C_F\right) a_{\mathrm{PS}}^{(1)}\, a^{(0)*} + C_F T_R \left(\frac{3}{2\epsilon} + \frac{3}{4}\right) a^{(0)}\, a^{(0)*}\right]$
$q\bar{q} \to Vg$, $qg \to Vq$	RVV 1L×1L	$v_q^2 + a_q^2$	$v^{(1)}\, v^{(1)*} - \frac{\beta_0}{2\epsilon}\, 2\mathrm{Re}\left[v^{(1)}\, v^{(0)*}\right] + \frac{\beta_0^2}{4\epsilon^2}\, v^{(0)}\, v^{(0)*}$
		$v_q \sum v_f$	$= 0$
		$a_q \sum a_f$ (L)	$2\mathrm{Re}\left[a_{\mathrm{PS}}^{(1)}\, a^{(1)*} - \left(\frac{\beta_0}{2\epsilon} + 2C_F\right) a_{\mathrm{PS}}^{(1)}\, a^{(0)*}\right]$
		$(\sum a_f)^2$ (L)	$a_{\mathrm{PS}}^{(1)}\, a_{\mathrm{PS}}^{(1)*}$
$gg \to Vg$	1L×1L	$(\sum v_f)^2$	$v_{\mathrm{PS}}^{(1)}\, v_{\mathrm{PS}}^{(1)*}$ (colour $d^{abc}d^{abc}$)
		$(\sum a_f)^2$ (L)	$a_{\mathrm{PS}}^{(1)}\, a_{\mathrm{PS}}^{(1)*}$
$q\bar{q} \to V$	VVV		three-loop form factor: not covered here

3 Parity-odd pieces (L_6)

Only for observables that resolve the lepton direction. Every row is Larin. Here Z_a^{ns} enters up to two loops: its a^2 coefficient $\frac{11}{2}C_F^2 - \frac{107}{36}C_FC_A + \frac{1}{18}C_Fn_f + \frac{C_F\beta_0}{\epsilon}$ is written out where needed. "$= 0$": see section 4.

Table 2: Parity-odd pieces, contracted with L_6 (all Larin).

channels	contribution	coupling	renormalised
LO (a^0)			
$q\bar{q} \to V$	B	$v_q a_q$	$2\mathrm{Re}\left[v^{(0)}\, a^{(0)*}\right]$
NLO (a^1)			
$q\bar{q} \to Vg$, $qg \to Vq$	R	$v_q a_q$	$2\mathrm{Re}\left[v^{(0)}\, a^{(0)*}\right]$
$q\bar{q} \to V$	V	$v_q a_q$	$2\mathrm{Re}\left[v^{(1)}\, a^{(0)*}\right] + 2\mathrm{Re}\left[a^{(1)}\, v^{(0)*} - 2C_F\, a^{(0)}\, v^{(0)*}\right]$
NNLO (a^2)			
$q\bar{q} \to Vgg$, $qg \to Vqg$, $gg \to Vq\bar{q}$	RR	$v_q a_q$	$2\mathrm{Re}\left[v^{(0)}\, a^{(0)*}\right]$
$q\bar{q} \to VQ\bar{Q}$	RR	$v_q a_q$	$2\mathrm{Re}\left[v^{(0)}\, a^{(0)*}\right]$
$qQ \to VqQ$	RR	$v_Q a_Q$	$2\mathrm{Re}\left[v_2^{(0)}\, a_2^{(0)*}\right]$
$q\bar{Q} \to Vq\bar{Q}$	RR	$v_q a_Q$	$2\mathrm{Re}\left[v^{(0)}\, a_2^{(0)*}\right]$
$q\bar{q} \to Vq\bar{q}$	RR	$a_q v_Q$	$2\mathrm{Re}\left[a^{(0)}\, v_2^{(0)*}\right]$
$qq \to Vqq$	RR		
$q\bar{q} \to Vg$, $qg \to Vq$	RV	$v_q a_q$	$2\mathrm{Re}\left[v^{(1)}\, a^{(0)*} - \frac{\beta_0}{2\epsilon}\, v^{(0)}\, a^{(0)*}\right] + 2\mathrm{Re}\left[a^{(1)}\, v^{(0)*} - \left(\frac{\beta_0}{2\epsilon} + 2C_F\right) a^{(0)}\, v^{(0)*}\right]$
	RV	$v_q \sum a_f$	$2\mathrm{Re}\left[a_{\mathrm{PS}}^{(1)}\, v^{(0)*}\right]$
	RV	$a_q \sum v_f$	$= 0$

Table 2: Parity-odd pieces, contracted with L_6 (all Larin).

channels	contribution	coupling	renormalised
$q\bar{q} \to V$	VV	$v_q a_q$	$2\mathrm{Re}\left[v^{(2)}\, a^{(0)*} - \frac{\beta_0}{\epsilon}\, v^{(1)}\, a^{(0)*}\right]$ $+ 2\mathrm{Re}\left[a^{(2)}\, v^{(0)*} - \left(\frac{\beta_0}{\epsilon} + 2C_F\right) a^{(1)}\, v^{(0)*} + \left(\frac{11}{2}C_F^2 - \frac{107}{36}C_F C_A + \frac{1}{18}C_F n_f + \frac{C_F\beta_0}{\epsilon}\right) a^{(0)}\, v^{(0)*}\right]$
	2L×0L	$v_q \sum a_f$	$2\mathrm{Re}\left[a^{(2)}_{\mathrm{PS}}\, v^{(0)*} + C_F T_R\left(\frac{3}{2\epsilon} + \frac{3}{4}\right) a^{(0)}\, v^{(0)*}\right]$
		$a_q \sum v_f$	$= 0$
$q\bar{q} \to V$	VV	$v_q a_q$	$2\mathrm{Re}\left[v^{(1)}\, a^{(1)*} - 2C_F\, v^{(1)}\, a^{(0)*}\right]$
	1L×1L		
N^{3}LO (a^3)			
$q\bar{q} \to Vggg$	RRR	$v_q a_q$	$2\mathrm{Re}\left[v^{(0)}\, a^{(0)*}\right]$
$qg \to Vqgg$			
$gg \to Vq\bar{q}g$			
$q\bar{q} \to VQ\bar{Q}g$	RRR	$v_q a_q$	$2\mathrm{Re}\left[v^{(0)}\, a^{(0)*}\right]$
$qg \to VqQ\bar{Q}$		$v_Q a_Q$	$2\mathrm{Re}\left[v_2^{(0)}\, a_2^{(0)*}\right]$
$qQ \to VqQg$		$v_q a_Q$	$2\mathrm{Re}\left[v^{(0)}\, a_2^{(0)*}\right]$
$q\bar{Q} \to Vq\bar{Q}g$		$a_q v_Q$	$2\mathrm{Re}\left[a^{(0)}\, v_2^{(0)*}\right]$
$q\bar{q} \to Vq\bar{q}g$			
$qq \to Vqqg$			
$qg \to Vqq\bar{q}$			
$q\bar{q} \to Vgg$	RRV	$v_q a_q$	$2\mathrm{Re}\left[v^{(1)}\, a^{(0)*} - \frac{\beta_0}{\epsilon}\, v^{(0)}\, a^{(0)*}\right] + 2\mathrm{Re}\left[a^{(1)}\, v^{(0)*} - \left(\frac{\beta_0}{\epsilon} + 2C_F\right) a^{(0)}\, v^{(0)*}\right]$
$qg \to Vqg$		$v_q \sum a_f$	$2\mathrm{Re}\left[a^{(1)}_{\mathrm{PS}}\, v^{(0)*}\right]$
$gg \to Vq\bar{q}$		$a_q \sum v_f$	$2\mathrm{Re}\left[v^{(1)}_{\mathrm{PS}}\, a^{(0)*}\right]$

Table 2: Parity-odd pieces, contracted with L_6 (all Larin).

channels	contribution	coupling	renormalised
$q\bar{q} \to VQ\bar{Q}$, $qQ \to VqQ$, $q\bar{Q} \to Vq\bar{Q}$, $q\bar{q} \to Vq\bar{q}$, $qq \to Vqq$	RRV	$v_q a_q$	$2\mathrm{Re}\left[v^{(1)}\, a^{(0)*} - \frac{\beta_0}{\epsilon}\, v^{(0)}\, a^{(0)*}\right] + 2\mathrm{Re}\left[a^{(1)}\, v^{(0)*} - \left(\frac{\beta_0}{\epsilon} + 2C_F\right) a^{(0)}\, v^{(0)*}\right]$
		$v_Q a_Q$	$2\mathrm{Re}\left[v_2^{(1)}\, a_2^{(0)*} - \frac{\beta_0}{\epsilon}\, v_2^{(0)}\, a_2^{(0)*}\right] + 2\mathrm{Re}\left[a_2^{(1)}\, v_2^{(0)*} - \left(\frac{\beta_0}{\epsilon} + 2C_F\right) a_2^{(0)}\, v_2^{(0)*}\right]$
		$v_q a_Q$	$2\mathrm{Re}\left[v^{(1)}\, a_2^{(0)*} - \frac{\beta_0}{\epsilon}\, v^{(0)}\, a_2^{(0)*}\right] + 2\mathrm{Re}\left[a_2^{(1)}\, v^{(0)*} - \left(\frac{\beta_0}{\epsilon} + 2C_F\right) a_2^{(0)}\, v^{(0)*}\right]$
		$a_q v_Q$	$2\mathrm{Re}\left[a^{(1)}\, v_2^{(0)*} - \left(\frac{\beta_0}{\epsilon} + 2C_F\right) a^{(0)}\, v_2^{(0)*}\right] + 2\mathrm{Re}\left[v_2^{(1)}\, a^{(0)*} - \frac{\beta_0}{\epsilon}\, v_2^{(0)}\, a^{(0)*}\right]$
		$v_q \sum a_f$	$2\mathrm{Re}\left[a_{\mathrm{PS}}^{(1)}\, v^{(0)*}\right]$
		$v_Q \sum a_f$	$2\mathrm{Re}\left[a_{\mathrm{PS}}^{(1)}\, v_2^{(0)*}\right]$
		$a_q \sum v_f,\ a_Q \sum v_f$	$= 0$
$q\bar{q} \to Vg$, $qg \to Vq$	RVV	$v_q a_q$	$2\mathrm{Re}\left[v^{(2)}\, a^{(0)*} - \frac{3\beta_0}{2\epsilon}\, v^{(1)}\, a^{(0)*} + \left(\frac{3\beta_0^2}{8\epsilon^2} - \frac{\beta_1}{4\epsilon}\right) v^{(0)}\, a^{(0)*}\right]$
			$+ 2\mathrm{Re}\left[a^{(2)}\, v^{(0)*} - \left(\frac{3\beta_0}{2\epsilon} + 2C_F\right) a^{(1)}\, v^{(0)*} + \left(\frac{3\beta_0^2}{8\epsilon^2} - \frac{\beta_1}{4\epsilon} + \frac{2C_F\beta_0}{\epsilon} + \frac{11}{2}C_F^2 - \frac{107}{36}C_F C_A + \frac{1}{18}C_F n_f\right) a^{(0)}\, v^{(0)*}\right]$
	2L×0L	$v_q \sum a_f$	$2\mathrm{Re}\left[a_{\mathrm{PS}}^{(2)}\, v^{(0)*} - \left(\frac{3\beta_0}{2\epsilon} + 2C_F\right) a_{\mathrm{PS}}^{(1)}\, v^{(0)*} + C_F T_R \left(\frac{3}{2\epsilon} + \frac{3}{4}\right) a^{(0)}\, v^{(0)*}\right]$
		$a_q \sum v_f$	$2\mathrm{Re}\left[v_{\mathrm{PS}}^{(2)}\, a^{(0)*}\right]$
$q\bar{q} \to Vg$, $qg \to Vq$	RVV	$v_q a_q$	$2\mathrm{Re}\left[v^{(1)}\, a^{(1)*} - \left(\frac{\beta_0}{2\epsilon} + 2C_F\right) v^{(1)}\, a^{(0)*}\right]$
			$- 2\mathrm{Re}\left[\frac{\beta_0}{2\epsilon}\, a^{(1)}\, v^{(0)*} - \frac{\beta_0}{2\epsilon}\left(\frac{\beta_0}{2\epsilon} + 2C_F\right) v^{(0)}\, a^{(0)*}\right]$
	1L×1L	$v_q \sum a_f$	$2\mathrm{Re}\left[a_{\mathrm{PS}}^{(1)}\, v^{(1)*} - \frac{\beta_0}{2\epsilon}\, a_{\mathrm{PS}}^{(1)}\, v^{(0)*}\right]$
		$a_q \sum v_f,\ \sum v_f \sum a_f$	$= 0$
$gg \to Vg$	1L×1L	$\sum v_f \sum a_f$	$= 0$ (colour $d^{abc} f^{abc} = 0$)
$q\bar{q} \to V$	VVV		three-loop form factor: not covered here

4 Why the "$= 0$" entries vanish

All of them are PS pieces: V on a closed quark loop, with the loop attached to the rest of the diagram by gluons.

Furry's theorem (charge conjugation). For every loop orientation there is the diagram with the opposite orientation. With $C\gamma^\mu C^{-1} = -(\gamma^\mu)^T$ and $C\gamma^\mu\gamma_5 C^{-1} = +(\gamma^\mu\gamma_5)^T$, the reversed loop equals the original one times $(-1)^{n_v}$, n_v = number of *vector* vertices on the loop (gluons and V if vector), with the colour trace read in the opposite order. For V plus two gluons the colour trace is $\mathrm{Tr}(T^aT^b) = T_R\delta^{ab}$, the same in both orders, so

- vector V: $n_v = 3$, the two orientations cancel: $v_{\rm PS} = 0$;
- axial V: $n_v = 2$, they add: $a_{\rm PS} \neq 0$ (the anomalous triangle).

Two gluons on the loop is the case of: (i) V + one parton at one loop ($q\bar q \to Vg$, $qg \to Vq$: one gluon to the quark line, one external); (ii) $q\bar q \to V$ at two loops (both gluons to the quark line); (iii) four-quark channels at one loop (one gluon to each quark line). Hence the zeros:

entry	vanishing amplitude	case
RV $v_q\sum v_f$; RV $a_q\sum v_f$ (odd)	$v^{(1)}_{\rm PS}$, $m=1$	(i)
RVV 1L×1L $v_q\sum v_f$, $a_q\sum v_f$, $\sum v_f\sum a_f$, $(\sum v_f)^2$	$v^{(1)}_{\rm PS}$, $m=1$	(i)
VV $v_q\sum v_f$; VV $a_q\sum v_f$ (odd)	$v^{(2)}_{\rm PS}$, $m=0$	(ii)
RRV four-quark $v_\ell\sum v_f$, $a_\ell\sum v_f$	$v^{(1)}_{\rm PS}$, four-quark	(iii)

With *three* gluons on the loop the vector part survives: $n_v = 4$, and the two orders of the trace add, $\mathrm{Tr}(T^aT^bT^c) + \mathrm{Tr}(T^cT^bT^a) = \frac{1}{2}d^{abc}$, while for the axial part they subtract, $\propto if^{abc}$. That is why $v^{(2)}_{\rm PS}$ for $m=1$, $v^{(1)}_{\rm PS}$ for $q\bar q \to Vgg$, $qg \to Vqg$, $gg \to Vq\bar q$, and $gg \to Vg$ are non-zero.

Colour. $gg \to Vg$ at one loop: vector $\propto d^{abc}$, axial $\propto f^{abc}$ (previous paragraph), so the parity-odd interference $v^{(1)}_{\rm PS}a^{(1)*}_{\rm PS} \propto d^{abc}f^{abc} = 0$ (symmetric times antisymmetric).

Reading the tables.

- Counterterms are lower-loop pieces *of the same labels*. So some Larin lower-order pieces are needed even though the lower order itself uses $aa = vv$: RRV a_qa_Q needs RR $a^{(0)}\,a_2^{(0)*}$ (L); RVV $a_q\sum a_f$ needs RV $a^{(1)}_{\rm PS}\,a^{(0)*}$ and R $a^{(0)}\,a^{(0)*}$ (L), and VV $a_q\sum a_f$ needs B $a^{(0)}\,a^{(0)*}$ (L) (the $1/\epsilon$ of the PS counterterm times Larin's $aa \neq vv$ at $O(\epsilon)$ gives a finite difference).
- $-2C_F$ and the a^2 term above are Larin's non-singlet renormalisation Z_a^{ns} of the axial vertex; $C_FT_R(\frac{3}{2\epsilon} + \frac{3}{4})$ is the PS constant Z_s (per flavour). The two-loop Z_a^{ns} appears only in the parity-odd table: the parity-even same-line products use $aa = vv$.
- With $n_f = 5$ massless flavours $\sum_f a_f \neq 0$ (the b has no partner): the $\sum a_f$ rows are the "b remainder" and need a decision on the top (`Larin.pdf`, sect. 5).
- In the parity-odd table each coupling has two orderings (vector in the amplitude, axial in the conjugate, and the reverse); the counterterm of each ordering carries the Z of the amplitude label at two loops, of the conjugate label in 1L×1L.
- Identical quarks ($q = Q$): same rows with $v_Q = v_q$, $a_Q = a_q$; pieces kept separate by line.

Origin of the coefficients. The amplitude with m emissions is $\bar a^{\,m/2}\big(M^{(0)} + \bar a M^{(1)} + \bar a^2 M^{(2)}\big)$ with $\bar a = a\big[1 - \frac{\beta_0}{\epsilon}a + \big(\frac{\beta_0^2}{\epsilon^2} - \frac{\beta_1}{2\epsilon}\big)a^2\big]$, times Z_a^{ns} for each Larin axial vertex, plus Z_s times the tree with V on the open lines for $a_{\rm PS}$. Expanding in a and keeping the stated order gives every row; the 1L×1L row is the square of the renormalised one-loop amplitude, the 2L×0L row the renormalised two-loop amplitude times the tree. Check: for the vector, the two RVV rows together give $-\frac{2\beta_0}{\epsilon}\,2\mathrm{Re}\,[v^{(1)}\,v^{(0)*}] + \big(\frac{\beta_0^2}{\epsilon^2} - \frac{\beta_1}{2\epsilon}\big)v^{(0)}\,v^{(0)*}$, which is the expansion of $\bar a^{\,1}$ and $\bar a^{\,2}$ in the bare $|M|^2$; the parity-odd v_qa_q rows

together give in addition one power of Z_a^{ns}. The PS rows reproduce arXiv:2306.10170 eq. (5.14), the Larin axial one eq. (5.13) (with $2C_F\beta_0/\epsilon$; the paper prints $2\beta_0/\epsilon$).